

Home Paper Submission-IV

submitted to the

INSTITUTE OF CHEMICAL TECHNOLOGY, MUMBAI

for partial fulfillment of the degree of

Bachelor of Chemical Engineering

(Semester VII)

Design of a plant to produce 20,000 TPA of 2-hydroxyethyl methacrylate

by

NIDHI KULKARNI

22CHE142



Institute of Chemical Technology, Mumbai

(University under Section 3 of UGC Act 1956;

Elite Status and Centre of Excellence, Government of Maharashtra)

Maharashtra, India

October 2025

VGG

Table of Contents

Chapter 1: Introduction.....	7
1.1 Product definition and properties.....	7
.....	8
1.2 Hazards and Toxicity (Sigma-Aldrich, 2006).....	8
1.3 Uses of HEMA.....	9
1.3.1 Industrial Applications.....	9
1.3.2 Biomedical Applications.....	9
1.4 Market Analysis.....	10
1.5 Capacity of the HEMA manufacturing.....	11
Chapter 2: Process selection.....	12
2.1 Literature survey.....	12
2.1.1 Epoxidation of Methacrylic Acid.....	12
2.1.2. Trans-esterification of Ethylene Glycol.....	12
2.2 Comparison of the two processes.....	13
2.2.1 Analysis of the Esterification reaction (Changzhou Tianma group, 2012).....	14
2.2.2 Analysis of Epoxidation of Methacrylic acid (Changzhou Hickory Chemistry Ltd. 2016). ..	15
2.2.3 Comparison of the Routes based on economic potential.....	15
2.2.4 Final selection of the process.....	17
2.3 Production of Methacrylic acid.....	17
2.3.1 C4 Oxidation Route.....	18
2.3.2 ACH (Acetone Cyanohydrin) Process.....	20
2.3.3 Oxidative Dehydrogenation / Dehydration of Isobutyric Acid (IBA).....	21
2.4 Comparison of the Three Processes.....	22
2.4.1 Analysis: C4 Oxidation Route.....	23
2.4.2 Analysis: ACH Process.....	24
2.4.3 Analysis: Isobutyric Acid (IBA) Oxidation.....	24
2.5 Feedstock Availability for Methacrylic Acid in India.....	24
2.6 Final Selection and Justification.....	25
Chapter 3: Site Selection.....	27
3.1: Factors for Site Selection.....	27
Chapter 4: Thermodynamic Feasibility.....	32
Chapter 5: Kinetics of the Reaction.....	34
5.1 Partial oxidation of Isobutene and Methacrolein.....	34
5.1.1 Kinetic parameters for partial oxidation of Isobutene to Methacrolein.....	34
5.1.2 Kinetics parameters for the partial oxidation of Methacrolein.....	35
5.2 Esterification of Methacrylic acid with Ethylene Oxide.....	36
5.3 Oxidation of Isobutene, Methacrolein and Methacrylic acid to side products.....	36
Chapter 6:.....	37
Reactor system material balance.....	41
Quencher.....	45
Extraction column.....	48
Chapter 7: Energy balance.....	52
References.....	56

Table of Figures

Figure 1: Structure of 2-Hydroxyethyl Methacrylate.....	7
Figure 2: Synthesis routes for HEMA.....	13
Figure 3: Partial oxidation of a C4 stream to give Methacrylic Acid.....	19
Figure 4: ACh process for Methacrylic acid production.....	21
Figure 5: Block Flow Diagram for syntheses of HEMA and MAA.....	26

Index of Tables

Table 1: Physico-chemical Properties of HEMA.....	8
Table 2: Imports of HEMA.....	11
Table 3: Comaprision of syntheses routes for HEMA.....	13
Table 4: Economic Potential of HEMA based on route using Ethylene Glycol.....	16
Table 5: Economic Potential of HEMA based on route using Ethylene Oxide.....	17
Table 6: Comparision of Syntheses routes for Methacrylic acid.....	22
Table 7: Site comparision.....	28
Table 8: Weightage of each parameter.....	29
Table 9: The grading of Sites Based on Parameters.....	29
Table 10: Weighted Average of Parameters.....	30
Table 11: Standard Gibbs energy of formation.....	32
Table 12: Gibbs energy of formation for the reactions.....	33
Table 13: Kinetic parameters for reaction 1, Benyahia et. al.....	34
Table 14: Kinetic parameters for reaction 2.....	35
Table 15: Kinetic parameters for reaction 3.....	36

Chapter 1: Introduction

2-Hydroxyethyl Methacrylate, abbreviated as **HEMA**, is a hydroxy ester of methacrylic acid. It is a monomer that is used to make hydrogels used in making UV cured coatings and adhesives. The polymer (poly Hydroxyethyl Methacrylate or **PHEMA**) is used to make hydrogels and contact lenses and are used for drug delivery in biomedical fields. Since the monomer polymerizes quite easily, hydroquinone or methoxyphenol are added as inhibitors.

1.1 Product definition and properties

Name: 2-Hydroxyethyl Methacrylate (HEMA)

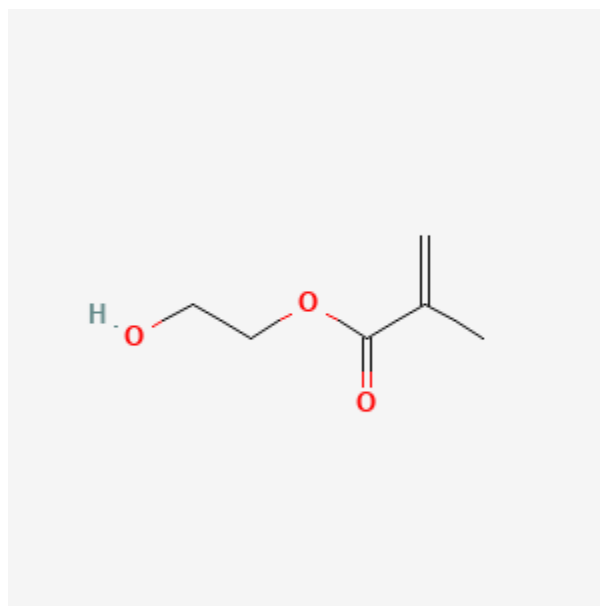


Figure 1: Structure of 2-Hydroxyethyl Methacrylate

Molecular formula: C₆H₁₀O₃

Appearance: Colourless, viscous liquid

CAS No: 868-77-9

IUPAC name: 2,2-hydroxy,ethyl-methacrylate

Other names: 2-hydroxyethyl 2-methylprop-2-enoate, Glycol methacrylate, and ethylene glycol methacrylate, Glycol mono-methacrylate

Table 1: Physico-chemical Properties of HEMA

Property	Value
Molecular Weight	130.14 g/mol
Density (@ 20°C)	1.07 g/cm ³
Melting point	-12 °C
Boiling point (@1 atm)	205 °C
Flash point (@1 atm)	107 °C
Solubility in water	Completely miscible
Vapor pressure (@ 20°C)	20 Pa

1.2

Hazards and Toxicity (Sigma-Aldrich, 2006)

Signal word: Warning.

Hazard statements: Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation.

Precautionary statements: Prevention: P261 Avoid breathing mist or vapors. P264 Wash skin thoroughly after handling. P280 Wear protective gloves/ eye protection/ face protection. Response: P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention. P337 + P313 If eye irritation persists: Get medical advice/ attention. P362 + P364 Take off contaminated clothing and wash it before re-using.

1.3 Uses of HEMA

The industrial and biomedical applications of HEMA (2-hydroxyethyl methacrylate) encompass a broad spectrum due to its versatile chemistry and properties. HEMA's high hydrophilicity and polymerizability make it indispensable in developing materials where mechanical strength, biocompatibility, and aqueous interaction are critical.

1.3.1 Industrial Applications

1. Synthesis of coating resins for automotive, architectural, and paper industries
2. Production of advanced adhesives and sealants
3. Textile finishing agents to improve fabric durability and water resistance
4. Printing inks for high-performance applications
5. Manufacture of hygiene products and plastic modifiers
6. Components in UV-curable and photopolymerizable formulations

1.3.2 Biomedical Applications

1. Contact lens hydrogels due to excellent water absorption and biocompatibility
2. Biosensors and membranes leveraging its inertness and hydrogel formation
3. Dental restorative materials such as adhesives and composite resins for hydrolytic and enzymatic stability
4. Controlled drug delivery systems using hydrogel networks for sustained release
5. Artificial organs and tissue engineering scaffolds
6. Cell culture substrates facilitating spheroid formation in biotechnology research

1.4 Market Analysis

Global production of 2-Hydroxyethyl Methacrylate (HEMA) is estimated to be around 300,000–350,000 tonnes per annum. The global market size in 2024 is valued at approximately USD 1.2 billion, with a compound annual growth rate (CAGR) of 6.5% for the forecast period 2024–2030. HEMA demand is primarily driven by its use in adhesives, coatings, resins, UV-curable materials, and contact-lens formulations.

According to trade statistics and industry reports, India currently has no domestic manufacturers of HEMA, and the country is entirely import-dependent. Imports during 2023–2024 are estimated at 7,000–8,000 tonnes per annum, mainly sourced from China, Japan, and Europe. Based on average import prices ranging between USD 2.5–10.5 per kg, the annual import value is estimated between USD 18 million and USD 84 million. There is negligible export activity from India, as all imported volumes are consumed domestically for downstream resin and coating applications. The exports are a very negligible quantity.

Globally, the largest manufacturer is Wanhua Chemical Group, operating a plant with a capacity of 39,500 tonnes per annum in China. Other significant producers include Dow Chemical Company (USA) and Mitsubishi Chemical Corporation (Japan). Smaller contributions come from Evonik Industries, Nippon Shokubai, and GEO Specialty Chemicals.

Due to the absence of indigenous manufacturing, the Indian HEMA market is driven by imports, meeting demand from local resin, paint, and polymer industries. Given the current market size and a healthy CAGR, India represents a potential opportunity for setting up a 10,000–40,000 TPA domestic production facility. Such a plant could substitute imports, reduce foreign exchange outflow, and meet the growing needs of coatings and specialty polymers sectors.

The price of HEMA varies significantly based on purity and grade. Technical-grade HEMA (used in coatings and adhesives) trades at around USD 2.5–4.0 per kg, whereas high-purity grades (for biomedical and optical applications) can exceed USD 10 per kg. Prices are sensitive to fluctuations in feedstock costs of methacrylic acid and ethylene oxide, as well as logistics and polymerization inhibitor requirements.

In conclusion, the Indian HEMA market is poised for steady growth in line with global trends. With increasing demand for adhesives, coatings, and advanced polymer materials, establishing domestic HEMA production presents a viable opportunity. This could position India as a regional supplier in South and Southeast Asia in the long term.

Table 2: Imports of HEMA

Quantity (MT)	Total value (million USD)	Value per kg	Country
444.54	70.75	160.00	Singapore
398.78	61.4	153.00	China
119.63	19.54	163.0	Japan
4.8	2.26	471.0	Switzerland
0.2	0.15	745.0	South Korea

1.5 Capacity of the HEMA manufacturing

The proposed capacity for the new 2-Hydroxyethyl Methacrylate (HEMA) plant is set at **20,000 tonnes per annum (TPA)**. At present, India imports approximately 7,000–8,000 tonnes of HEMA every year, as there are no indigenous producers. Since the estimated CAGR for HEMA is 6.5%, setting up a plant with capacity of 20,000 TPA, which is approximately half of the largest producer's capacity, the plant can be economically feasible and it will be easy to sustain the market.

Chapter 2: Process selection

2.1 Literature survey

Upon doing a literature survey, it was found that two major industrial processes are used to produce HEMA economically. The first mention of the synthesis is dated to 1925. The two common methods of synthesis are (i) Reaction of Methacrylic acid with Ethylene Glycol and (ii) Reaction of Methacrylic acid with Ethylene Oxide (). Both reactions can lead to formation of Ethylene Glycol Di-methacrylate (abbreviated as **EDGMA**). Other reactions such as acyl chloride esterification method and dihydroxyl carbonic esterification reactions are also possible reactions to produce HEMA, however the conditions required for these reactions are achievable only at lab scale.

2.1.1 Epoxidation of Methacrylic Acid

Upon reacting ethylene oxide with methacrylic acid, the reaction producing HEMA proceeds by a nucleophilic substitution reaction at the (alpha) carbon. The reactivity of ethylene oxide plays a significant role in designing the process since it is a highly reactive and toxic compound. Hence the reaction is conducted at controlled temperatures and pressures. The common choices for a solvent media are any tertiary amines.

2.1.2. Trans-esterification of Ethylene Glycol

An alternative route involves the esterification of methacrylic acid with ethylene glycol. In this process, methacrylic acid reacts with an excess of ethylene glycol in the presence of an acid catalyst—commonly sulfuric acid, p-toluene sulfonic acid, or an acidic ion-exchange resin. The reaction proceeds via condensation, forming 2-hydroxyethyl methacrylate and water as a by-product. The formation of water is the major drawback of this method, since HEMA is completely miscible in water. Water also acts as a polymerization promotor and hence it leads to the formation of hydrogel.

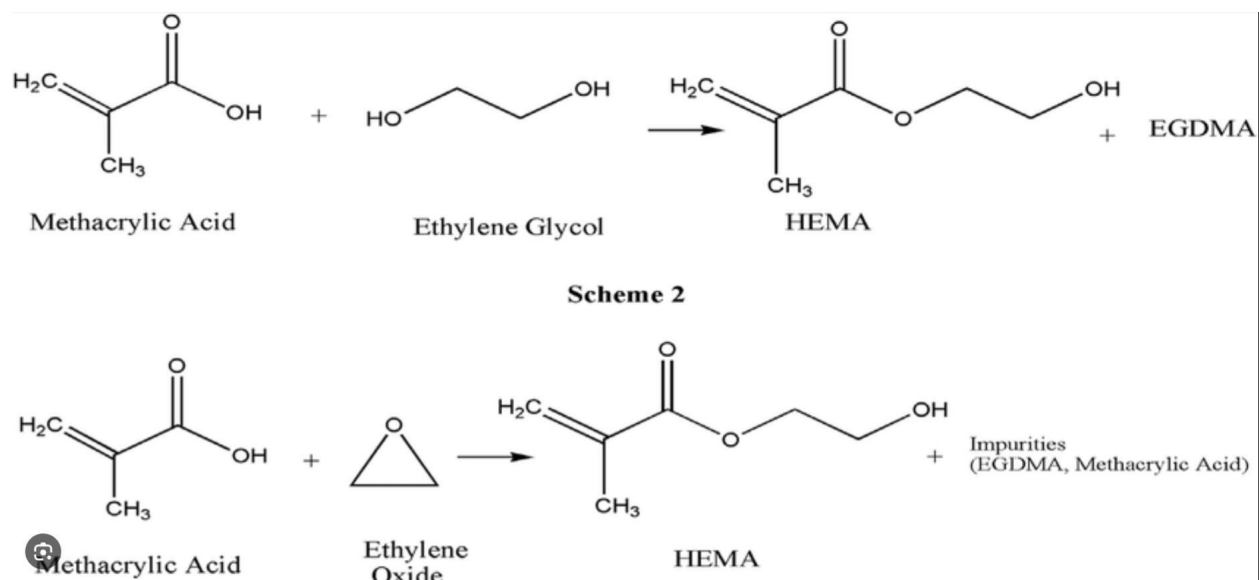


Figure 2: Synthesis routes for HEMA

2.2 Comparison of the two processes

Table 3: Comparison of synthesis routes for HEMA

Process Variable	Esterification Reaction	Epoxidation Reaction
Pressure	Atmospheric Pressures	Vacuum of -0.99 – -0.75 Bar
Temperature	100-120 °C	75 -90 °C
Catalyst	P-methyl benzene sulfonic acid or ion exchange catalysts	Ferric Formate or Chromium Acetate
Reagent	Sulphuric acid	Tertiary amine such as pyridine, N,N-Dimethyl toluene etc.
Reaction time	1 – 3 hr	1.5 – 2 hr

Percentage yield	> 85%	> 99%
Polymerization inhibitor	Phenothiazine	MEHQ (hydroquinone monomethyl ether)
Impurities	Ethylene glycol dimethacrylate, Water	Ethylene glycol dimethacrylate

2.2.1 Analysis of the Esterification reaction (Changzhou Tianma group, 2012)

Advantages of the reaction are as follows-

1. The reaction is relatively simpler and it does not require using complex and hazardous raw materials, which also makes it safer than the ring opening reaction.
2. The reaction conditions are moderate.
3. There is no need to use expensive metal oxide catalysts.
4. Since ion-exchange resins can be used, the separation can be done much more easily.

Disadvantages of the reaction are as follows-

1. The reaction is limited by equilibrium.
2. The amount of side product (EGDMA) produced is much higher even if the amount of acid is controlled. There is a high change of a co-polymer formation if the amount of EGDMA is not controlled.
3. The water formed in this reaction is a major issue since HEMA is highly soluble in water. Water also promotes the polymerization reaction.
4. The yield and selectivity is also lower for this reaction

2.2.2 Analysis of Epoxidation of Methacrylic acid (Changzhou Hickory Chemistry Ltd. 2016)

Advantages of the reaction are as follows-

1. The reaction has better yield and selectivity and the amount of side-product is less.
2. There is no water formed as it is a direct one step synthesis.
3. The purification hence becomes simpler.

Disadvantages of the reaction are as follows-

1. Ethylene oxide is a highly flammable, highly carcinogenic and reactive material.
2. The reaction requires a very controlled environment. If the pressures or temperatures deviate even a little, there is a very high chance of a runaway reaction and a very major accident.

2.2.3 Comparison of the Routes based on economic potential

Economic potential is essentially the net difference between the price of the raw materials and the product. It can be used to compare the economic feasibility of two reactions to synthesize a chemical product.

The economic potential (EP) of a chemical product is given by the following formula

EP = Value of Products – Cost of Raw Materials and Utilities

or equivalently,

$$EP = \Sigma(m_p \times P_p) - [\Sigma(m_r \times P_r)]$$

Where-

m_p is the Mass (or molar) flow rate of product p

P_p is the Price of product p (Rs/kg or \$/kg)

m_r is the Mass (or molar) flow rate of raw material r

P_r is the Price of raw material r

Using this, we can calculate the economic potential of the two routes. The route with more EP will be more feasible.

Table 4: Economic Potential of HEMA based on route using Ethylene Glycol

Chemical	Moles	MW	Price/kg	price/mole
EG	1	62	65	4.03
MAA	1	86	150	12.9
H2O	1	18	0	0
HEMA	1	130	160	20.8

For the esterification reaction,

$$EP = 20.8 - 12.9 - 4.03$$

$$EP = 3.87$$

For the esterification reaction,

$$EP = 20.8-12.9-4.03$$

$$EP = 3.87$$

Table 5: Economic Potential of HEMA based on route using Ethylene Oxide

Chemical	Moles	MW	Price/kg	price/mole
EO	1	44	86	3.784
MAA	1	86	150	12.9
HEMA	1	130	160	20.8

For the epoxidation reaction,

$$EP = 20.8-12.9-3.78$$

$$EP = 4.12$$

2.2.4 Final selection of the process

Based on the factors such as economic potential, raw materials used, yield, selectivity and the reaction conditions, the epoxidation route is more feasible for bulk production of HEMA. This route is also preferred globally.

2.3 Production of Methacrylic acid

Setting up a methacrylic acid (MAA) manufacturing plant alongside a 2-hydroxyethyl methacrylate (HEMA) plant is necessary for both economic and technical reasons. Economically, methacrylic acid is a key raw material for HEMA production through epoxidation; producing MAA onsite reduces raw material procurement costs, transportation expenses, and supply chain risks associated with market fluctuations or delays. This integration improves overall cost efficiency by enabling economies of scale in raw material handling, utilities, and infrastructure usage. Technically, the close coupling of methacrylic acid production with HEMA manufacturing streamlines process flow, enhances material purity control, and minimizes losses during intermediate transfer. Additionally, onsite MAA production provides flexibility in adjusting HEMA feed composition and scaling production volumes rapidly based on market demand. Together, these factors support a more robust, cost-effective, and technically optimized specialty monomer manufacturing facility essential for maintaining competitiveness in rapidly growing biomedical, coatings, and industrial segments where HEMA is widely employed.

A literature survey shows three major industrially relevant routes for producing methacrylic acid (MAA):

1. C4 oxidation (oxidation of C4 hydrocarbons such as isobutene/isobutane to methacrolein/methacrylic acid)
2. ACH (Acetone Cyanohydrin) process (conversion of acetone via acetone cyanohydrin to methacrylamide and then to MAA/MMA)
3. Oxidative dehydrogenation/dehydration of isobutyric acid (IBA) to MAA.

All three routes have been studied and commercialised to varying extents. The ACH route has been the historical commercial route for MMA/MAA, while C4 oxidation routes (via isobutene → methacrolein → MAA) and IBA oxidative dehydrogenation/dehydration have been investigated and implemented depending on feedstock availability and economics.

2.3.1 C4 Oxidation Route

The C4 oxidation route typically oxidises C4 streams (isobutene, isobutane, or mixtures from steam

crackers) to methacrolein (MAL) and subsequently to methacrylic acid (MAA). It is a controlled catalytic oxidation of the C4 rich raw material. (Roma Chemical Ltd, 2006)

Typical catalysts & conditions

1. Catalysts: vanadium-phosphorus based catalysts (e.g., VPO, vanadium pyrophosphate), mixed metal oxides (Mo–V–P, V–Mo–Nb), and supported vanadium oxides.
2. Reaction steps: partial oxidation of isobutene/isobutane to methacrolein (gas-phase, oxidative catalysts), followed by oxidation of methacrolein to MAA.
3. Temperatures: commonly in the range ~300–450 °C for gas-phase oxidation catalysts for direct IBA oxidation; partial oxidation steps for C4 → methacrolein operate at lower temperatures (200–350 °C) depending on catalyst.
4. Pressure: near atmospheric to a few bar depending on reactor design.

Yields & selectivity

Industrial C4 oxidation processes can achieve high conversions with varying selectivity to MAA/MAL depending on catalyst and process optimisation. Reported selectivities to desired C4 oxidation products (MAL/MAA) are high but side oxidation to CO_x and other by-products occurs and must be minimized with appropriate catalyst and reactor design.

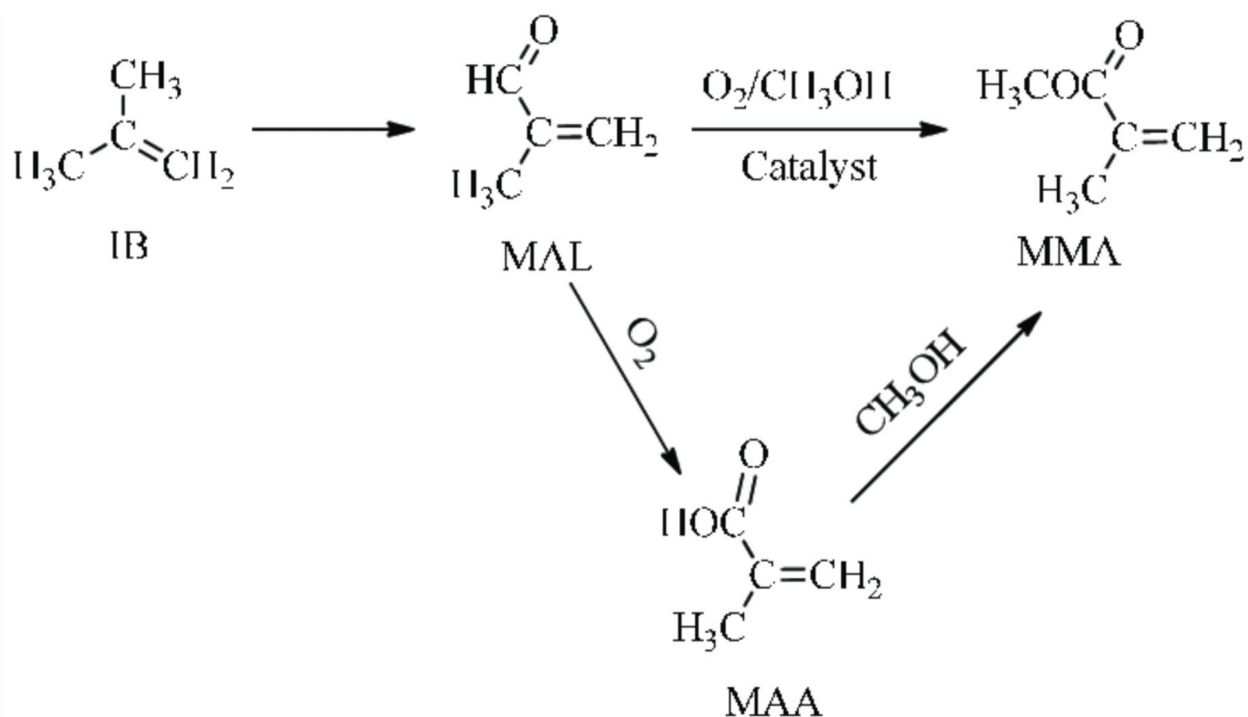


Figure 3: Partial oxidation of a C4 stream to give Methacrylic Acid

2.3.2 ACH (Acetone Cyanohydrin) Process

The ACH process converts acetone to acetone cyanohydrin (ACH) by reaction with hydrogen cyanide (HCN). ACH is then converted (hydrolysis followed by further steps) to methacrylamide or methacrylic derivatives and ultimately to MAA or MMA. Historically, the ACH route has been a major commercial route for methyl methacrylate (MMA) and related methacrylate chemicals. (Curtis, I. Carlson et. al., 2003)

Typical catalysts & conditions

1. ACH formation: acetone + HCN in the presence of a basic catalyst (phase transfer or base). Conditions are typically ambient to mild exothermic conditions; HCN handling and scrubbing systems are essential.
2. ACH hydrolysis / conversion: strongly acidic hydrolysis (traditionally sulfuric acid), or modern heterogeneous catalysts to avoid large quantities of waste acid. Temperatures and pressures vary with specific plant designs.

Yields & issues

The ACH route can give high yields to downstream methacrylic products when optimised. However, the process involves hydrogen cyanide—a highly toxic and hazardous reagent—and forms by-products such as acetone sulfonic acids, formamide (from HCN hydrolysis), and requires careful waste management.

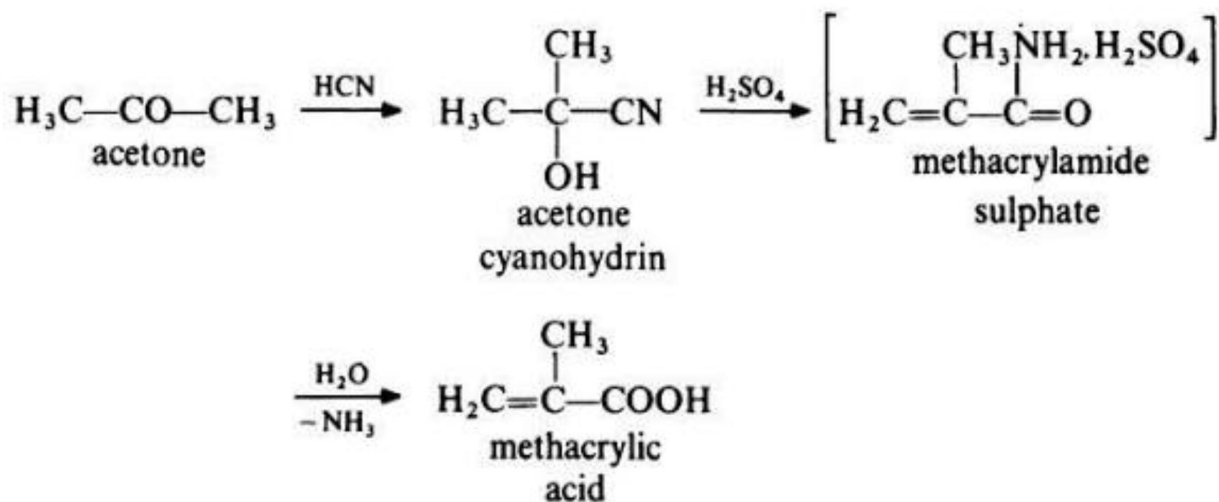


Figure 4: ACh process for Methacrylic acid production

2.3.3 Oxidative Dehydrogenation / Dehydration of Isobutyric Acid (IBA)

Isobutyric acid (IBA) can be oxidatively dehydrogenated/dehydrated to methacrylic acid in the gas phase over selective oxidation catalysts.

Typical catalysts & conditions

1. Catalysts: molybdenum- and vanadium-containing mixed oxides, heteropoly acids supported on silica, and modified molybdophosphoric materials.
2. Temperatures: typically 300–450 °C for gas-phase oxidative dehydrogenation.
3. Pressure: atmospheric to a few bar.

Yields & selectivity

Reported selectivities for IBA → MAA are moderate—literature and patents report selectivities in the

range of roughly 60–75% under certain conditions. Side reactions and over-oxidation reduce overall selectivity and must be managed by catalyst choice and reaction conditions.

2.4 Comparison of the Three Processes

Table 6: Comparison of Syntheses routes for Methacrylic acid

Process Variable	C4 Oxidation	ACH Process (Acetone Cyanohydrin)	IBA Dehydration / Oxidative Dehydrogenation
Pressure	Near atmospheric – 3 bar	1–2 bar (ACH formation), hydrolysis may use slightly higher pressure	Atmospheric – 2 bar
Temperature	200–450 °C (stepwise for oxidation)	0–50 °C for ACH formation; 60–120 °C for hydrolysis	300–450 °C
Catalyst	Vanadium-phosphorus oxides, Mo–V–P mixed oxides, supported V oxides	Acid catalysts (sulfuric acid, strong solid acids)	Molybdenum-vanadium mixed oxides, heteropoly acids
Reagent	Crude C4 stream (isobutene/isobutane), air/oxygen	Acetone + HCN (hazardous)	Isobutyric acid, O ₂ / air for oxidative route
Reaction time	Seconds to minutes	Minutes to hours	Minutes (gas-phase)

	(continuous gas-phase)	depending on step	continuous)
Percentage yield	70–90% (to MAA, depending on selectivity)	80–95% (stoichiometric, high yield)	60–75% (moderate selectivity)
Impurities / By-products	CO, CO ₂ , maleic anhydride, polymeric tars, methacrolein residues	Acetone, ammonium salts, residual HCN, acetone sulfonic acid	CO _x , overoxidation products, polymeric residues

2.4.1 Analysis: C4 Oxidation Route

Advantages:

1. Compatible with crude C4 streams available at steam crackers (butenes, butadiene-rich cuts).
2. Can be configured as a continuous gas-phase oxidation process with high throughput.
3. Mature catalyst families (V–P, V–Mo, Mo–V–P) exist and have been optimised for selective partial oxidation steps.

Disadvantages and possible side reactions:

1. Over-oxidation to CO and CO₂ (complete combustion) reduces selectivity.
2. Formation of by-products such as maleic anhydride, crotonaldehyde, acetaldehyde, and polymeric tars depending on feed composition and conditions.
3. Partial oxidation may produce methacrolein (MAL) as an intermediate—controlling the two-step oxidation is critical to avoid accumulation and side reactions.
4. Catalyst deactivation (coking, phase changes) and need for regeneration.

5. Presence of butadiene and other dienes in crude C4 can lead to undesired reactions (polymerisation, condensation) and require upstream purification.

2.4.2 Analysis: ACH Process

Advantages:

1. Historically proven route to MMA/MAA with high potential yields when optimised.

Disadvantages:

1. Requires hydrogen cyanide (HCN): highly toxic, poses acute safety hazards and complex environmental controls.
2. Side-products and acid wastes (if sulfuric acid used) can generate disposal and corrosion issues.
3. Modern improvements (heterogeneous catalysts for hydrolysis) reduce some waste but do not eliminate HCN hazards.

2.4.3 Analysis: Isobutyric Acid (IBA) Oxidation

Advantages:

1. Direct conversion from a C4 carboxylic acid; suitable where IBA feedstock is available.

Disadvantages:

1. Requires high temperatures (300–450 °C) and selective catalysts.
2. Selectivity to MAA is moderate (often 60–75%) with competing over-oxidation and by-product formation.

2.5 Feedstock Availability for Methacrylic Acid in India

Dahej is a major petrochemical hub with steam cracker and petrochemical complexes (OPaL/ONGC Petro additions and other operators) producing ethylene, propylene and C4 fractions as coproducts.

Crude C4 streams (containing isobutene, butenes, butadiene, and butanes) are commonly produced at crackers and can be sourced from cracker operators, refineries, and C4 traders in the Dahej PCPIR/Dahej industrial cluster. Proximity to these crackers makes Dahej a favourable site for C4 oxidation processes.

2.6 Final Selection and Justification

Based on catalyst options, yields, continuous-operation compatibility, and feedstock availability in petrochemical clusters such as Dahej, the **C4 oxidation route** is selected for bulk production of methacrylic acid.

1. The C4 oxidation route integrates well with steam cracker crude C4 availability at Dahej, minimising feedstock logistics and cost.
2. It can be operated continuously with well-studied oxidation catalysts to achieve economically favourable yields and throughput.
3. The ACH route was rejected primarily because it requires hydrogen cyanide (HCN), which is highly hazardous and requires stringent safety measures and complex waste handling.
4. IBA oxidation was not selected due to moderately lower selectivity and the requirement for high-temperature operation and specialised catalyst systems.
5. The global competitors for methacrylic acid are also shifting to the C4 oxidation route due to the advantages of this process. Hence if we want to establish a plant for the long run, it is better to go for the C4 route.

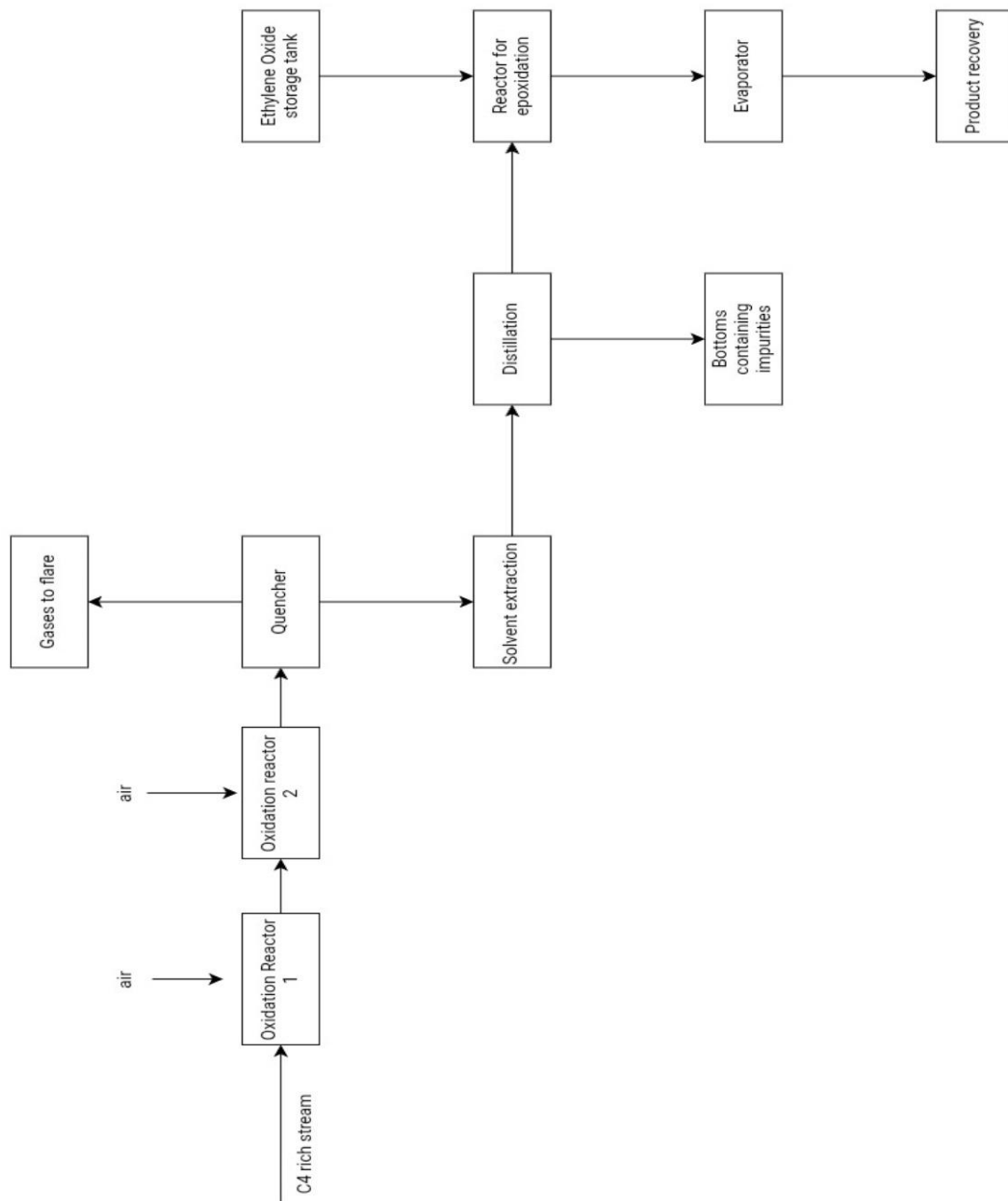


Figure 5: Block Flow Diagram for syntheses of HEMA and MAA

Chapter 3: Site Selection

3.1: Factors for Site Selection

For a methacrylic acid (MAA) and 2-hydroxyethyl methacrylate (HEMA) plant, site selection is a critical strategic decision due to the hazardous, bulk nature of raw materials and the multi-step chemical processes involved. The core criteria are:

- **Proximity to Petrochemical Feedstock:** The catalytic oxidation process for MAA requires a reliable supply of C4 hydrocarbons and oxygen. Siting the plant near large petrochemical complexes minimizes feedstock transport costs and ensures operational continuity. For HEMA, access to methacrylic acid is essential for the epoxidation step; co-location within an established methacrylate cluster enables efficient intermediate handling.
- **Logistics Infrastructure:** MAA and HEMA plants demand robust inbound and outbound transport networks (road, rail, port) to handle bulk chemical supply and enable timely export of finished products. Integrated clusters with efficient connectivity help minimize delays and logistics expenses.
- **Utilities and Steam Availability:** These processes are energy-intensive, needing reliable electricity and process steam. Clusters like Dahej and Vapi have existing utility networks, and proximity to large chemical sites allows for possible steam purchase rather than captive generation.
- **Water Supply and Effluent Management:** Large quantities of water are required for reaction, cooling, and cleaning. Sites must have a sustainable water source and advanced effluent treatment facilities due to the hazardous nature of organic acids and solvents.
- **Land and Regulatory Compliance:** Sufficient land for plant, expansion, safety buffer zones, and environmental approvals. Gujarat and Maharashtra offer streamlined chemical sector land acquisition and clearances.
- **Skilled Labor and Industrial Ecosystem:** Availability of trained chemical engineers and technicians, plus proximity to suppliers of catalysts, solvents, and equipment.
- **Ease of Doing Business:** Favorable regulations, industrial cluster support services, and ease of obtaining hazardous chemical permits all impact setup and ongoing operations.
- **Cluster Synergy:** Co-location with allied industries (such as resins, adhesives, paints) in clusters facilitates feedstock availability, shared services, and downstream integration.

Justification for Exclusion of Certain Parameters

- Proximity to end-user market is less critical: MAA/HEMA are shelf-stable, and Indian demand is spread nationally; clusters are connected to export ports.
- Climate is not a major factor as chemicals degrade minimally under standard storage, and clusters have adequate warehousing.
- Effluent processing is ensured by cluster-scale treatment facilities; individual plants do not need to independently construct ETPs unless mandated.
- Housing and labor supply are sufficient in semi-urban chemical hubs, and relocation/hiring logistics are straightforward.

3.2: Site Comparison and Selection with Justification

Table 7: Site comparison

Parameter	Dahej PCPIR, Gujarat	Ankleshwar, Gujarat	Vapi, Gujarat	Thane, Maharashtra
Petrochemical Complex	GNFC, Reliance, ONGC	GNFC, Narmada Chemicals	Diverse Chemicals, SME units	Deepak Fertilizers, legacy clusters
Transport Links	Major port, rail, road	Rail, road, moderate port	Rail, road, Mumbai port	Rail, road, Mumbai port
Utility Cost	Low, cluster tariffs	Low, cluster tariffs	Moderate, SME rates	Higher tariffs
Steam Availability	Cluster steam supply	Cluster steam, captive	Cluster steam, captive	Captive only
Water Cost (Rs/kL)	15–18	18–20	19–22	24–28
Land Cost (Lakh/acre)	16–22	13–20	15–24	38–45
Labor Cost	480	490	520	560

(Rs/day)				
-----------------	--	--	--	--

Table 8: Weightage of each parameter

Parameter	Weight (0-10)
Petrochemical complex	10
Transport links	9
Utility/steam	8
Water cost	7
Land cost	8
Labor cost	5
Ease of doing business	8
Cluster synergy	8

Table 9: The grading of Sites Based on Parameters

Parameter	Dahej	Ankleshwar	Vapi	Thane
Petrochemical complex	10	9	8	7
Transport links	10	8	8	8
Utility/steam	9	8	8	6
Water cost	9	8	8	6
Land cost	8	9	7	5

Labor cost	7	7	7	8
Ease of doing business	9	9	7	7
Cluster synergy	10	9	9	8

Table 10: Weighted Average of Parameters

Weighted Parameter	Dahej	Ankleshwar	Vapi	Thane
Petrochemical complex	100	90	80	70
Transport links	90	72	72	72
Utility/steam	72	64	64	48
Water cost	63	56	56	42
Land cost	64	72	56	40
Labor cost	35	35	35	40
Ease of doing business	72	72	56	56
Cluster synergy	80	72	72	64
Total weighted score (out of 10)	6.9	6.6	5.9	5.0

Based on the weighted score comparison, **Dahej PCPIR** in Gujarat emerges as the optimal location for setting up a methacrylic acid and 2-hydroxyethyl methacrylate plant. This site offers direct access to

petrochemical feedstocks, export ports, cluster steam/utilities, cost-effective land, and a mature chemical manufacturing ecosystem, making it ideal for specialty monomer production and expansion.

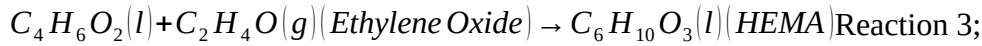
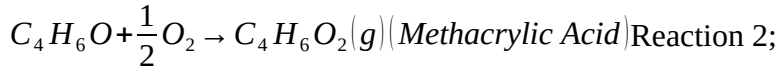
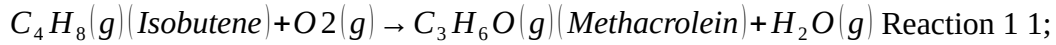
Chapter 4: Thermodynamic Feasibility

The thermodynamic feasibility of any process is characterized by the Gibbs free energy of that process at the given condition. The Gibbs free energy or ΔG for any process should be negative for the process to be feasible. The standard Gibbs free energy or $\Delta^\circ G$ is the ΔG at 25 °C or 298 K. The Table below lists the standard Gibbs energy of formation of the main species involved in the synthesis of HEMA using the chosen process. (All the thermodynamic values have been taken from Yaws C.L. , "Physical Properties Handbook", McGraw Hill publications)

Table 11: Standard Gibbs energy of formation

Species	$\Delta^\circ G$ formation (kJ/mol)
Isobutylene	58.07
Methacrolein	-57.6
Methacrylic Acid	-287.72 for gas phase, -203.65 for liquid phase
Water	-13.1
Ethylene Oxide	-291.81
2-Hydroxyethyl Methacrylate	-228.64

The reactions occurring for forming HEMA are as follows-



The ΔG° of any reaction is given by the formula

$$\Delta G^0 = \sum v_i G_{fi}$$

where v is the stoichiometric coefficient of species “i” and G_f is the Gibbs energy of formation for the species.

Using this relation, we obtain the Gibbs energy of formation at standard conditions for all the three major reactions. The results are in the table given below-

Table 12: Gibbs energy of formation for the reactions

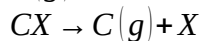
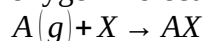
Reaction no.	Gibbs energy (kJ/mol)
1	-344.31
2	-230.12
3	-75.06

Since for all three reactions the ΔG° is negative, we conclude that all the reactions are thermodynamically feasible. Since the ΔG° is not zero, the reactions are not limited by equilibrium.

Chapter 5: Kinetics of the Reaction

5.1 Partial oxidation of Isobutene and Methacrolein

The partial oxidation of both Isobutene and Methacrolein follow the Mars-van Krevelen mechanism. The Mars-van Krevelen mechanism is when the hydrocarbon gas molecule reacts with the adsorbed oxygen molecule in the reactant pore. The mechanism is as follows-



where, A represents Oxygen, X represents the vacant site on the catalyst, B represents the hydrocarbon species and C is the product molecule. The reaction has a characteristic factor of ϕ which the desorption rate of Oxygen from the catalyst site.

5.1.1 Kinetic parameters for partial oxidation of Isobutene to Methacrolein

Molybdenum based catalysts provide the highest selectivity for this reaction. Based on the work done by Hoornaerts et. Al in their 1997 paper titled “Modification of kinetic parameters by action of oxygen spillover in selective oxidation of isobutene to methacrolein” the rate equation for this reaction is given as

$$r_{MAC} = \left(\frac{k_{red} P_{C4}^x k_{ox} P_{O2}^y}{k_{red} P_{C4}^x + k_{ox} P_{O2}^y} \right)$$

The values of the kinetic parameters are given in Table 13.

Table 13: Kinetic parameters for reaction 1, Benyahia et. al

Temperature (K)	k_red	k_ox	X	Y	ψ
	(mol atm ⁻¹ s ⁻¹ m ⁻²)	(mol atm ⁻¹ s ⁻¹ m ⁻²)			
623	8.88e-11	0.92e-11	0.03	0.7	0.16

643	16.82e-11	1.72e-11	0.25	0.77	0.09
663	26.6e-11	3.47e-11	0.33	0.7	0.12

The activation energy for catalyst reduction is 52 kJ/mol and for reoxidation is 142 kJ/mol. The constant K0 for the kinetic parameter is chosen to the parameters at 643 K due to the low psi value and Arrhenius equation is used to calculate the values at any temperature.

5.1.2 Kinetics parameters for the partial oxidation of Methacrolein

Heteropoly acid catalysts are used for facilitating these reactions. They are usually phospho-Molybdate complexes. The rate equation for the oxidation of Methacrolein is given by

$$-r = \frac{K_{O_2} P_{O_2}^n (1 + P_{H_2O}^m)^2}{K_{O_2} P_{O_2}^n (1 + P_{H_2O}^m) + 0.5 K P_{MAC}}$$

where, n = 0.7, m = 0.65 and the values of the kinetic parameters are described in the following table

Table 14: Kinetic parameters for reaction 2

Kinetic Parameter	Value (mol/hkgcatPa ^x)	Activation energy (kJ/mol)
Ko2	5.35E-06	99.99
K	7.6E-04	154.4

5.2 Esterification of Methacrylic acid with Ethylene Oxide

The rate equation for this reaction is given by

$$-r_{EO} = \frac{K_1 K_2 k_3 [EO][MA]}{1 + K_1 [EO] + K_1 K_2 [EO][MA] + K_4^{-1} [HEMA]}$$

where, [Cx] is the concentration of the species “x” and the value of the kinetic parameters are given in the table below.

Table 15: Kinetic parameters for reaction 3

Kinetic Parameter	Value at 50 °C	Value at 60 °C	Value at 70 °C	Value at 80 °C	Value at 90 °C	Activation energy (kJ/mol)
K1	6.654	3.909	2.368	1.478	0.944	33.53

K2	0.396	0.368	0.344	0.322	0.304	47.62
k3	0.0887	0.183	0.459	1.095	2.488	54.11
K4	0.08	0.19	0.428	0.921	1.901	77.21

The catalysts for this reaction include iron or chromium based catalysts. The use of a tertiary amine also helps in promoting this reaction.

5.3 Oxidation of Isobutene, Methacrolein and Methacrylic acid to side products

Chapter 6: Material Balance

Material balance is essentially applying the law of conservation of mass to the reactor system. We can do an overall balance for the reactor by using the known conversion values. The conversion of Isobutene in reactor 1 is 99.6% and the yield of Methacrolein is 79.6%. Similarly for reactor 2, the conversion of methacrolein is 78% and the selectivity is 85%. Low conversions of methacrolein are preferred since at conversions higher than 78%, the selectivity to Methacrylic acid decreases. For the esterification, the conversion with respect to methacrylic acid is 99.5%. Here, ethylene oxide is the limiting reactant due to its high reactivity.

For given production rate of 8,000 tonnes per annum, we have the production rate of HEMA to be 1 tonnes per hour or 19.23 kmol/h. Using this we back calculate the amount of Isobutene

required. The flow rate of methacrylic acid production will be approximately be the same. Using patent no. CN114057571A as a reference, we get the value of isobutylene molar flow rate as 41.75 kmol/h. The calculations done for this is as follows -

$$F_{\text{MAA}} = F_{\text{MAL}} \times \text{yield of MAA} = F_{\text{MAL}} \times 0.663$$

$$F_{\text{MAL}} = 33.23 \text{ kmol/h}$$

$$F_{\text{MAL}} = F_{\text{IB}} \times \text{yield of MAL} = F_{\text{IB}} \times 0.796$$

$$F_{\text{IB}} = 41.75 \text{ kmol/h.}$$

Where, F_x is the molar flow rate of the species “x”,

MAL is abbreviation of Methacrolein

MAA is abbreviation of Methacrylic acid.

Now from literature, it is seen that the ideal range for the molar ratio of IB to Oxygen to steam is 1: 2: 1.5. Using this composition, we can easily calculate the inlet flow rate to reactor R201 and its composition. This stream is labelled as stream 4 in the Process flow diagram.

Table 17: Stream composition of inlet to the first reactor

Stream 4	mol%	kmol/h	mw	kg/h
IB	8.63	41.75	56	2,338.00
O2	17.25	83.50	32	2,672.00
N2	61.18	296.05	28	8,289.27

steam	12.94	62.63	18	1,127.25
Total	100	483.92		14,426.52

The mass balance of the outlet stream from the reactor and its succeeding units has been carried out and found out based on the compositions mentioned in the patent CN114057571A. For the species whose molar composition in a stream was unknown, a basic elemental balance was used to find out the same. For each stream it was ensured that the mass in was equal to the mass out.

Given below is the list of the all the units in the plant

Table 18: List of the units in the plant

Equipment	Description	Type
R201	Oxidation reactor	Multi tubular packer bed
R202	Oxidation reactor	Multi tubular packer bed
C301	Quencher	Direct contact using water
C203	Crystallizer	Batch crystallizer using cooling water
Filter		To filter out TPA crystals
T204	Extraction column	Packed column
T302	Distillation column	Plate column
T303	Distillation column	Plate column
T304	Distillation column	packed

T204	Distillation column	packed
R305	Esterification reactor	Cstr (gas-liq)
T306	Distillation column	packed

6.1 Reactor system material balance

For the first oxidation reactor **R201**, the feed is Isobutene, Steam and air mixture which is initially heated to 350 °C and sent to the reactor. The reactor is a multi-tubular packed bed reactor which is said to operate isothermally. The coolant used is a thermic salt.

The conversion of Isobutene is 99.4% and the selectivity to methacrolein is 80%. Using these values, we can calculate the amount of Methacrolein that will be formed. Let molar flow rate of isobutene be denoted by F_{IB0} and the molar flow rate of methacrolein be F_{MAC} .

So,

$$F_{MAC} = 0.994 \times 0.8 \times F_{IB}$$

$$F_{MAC} = 0.994 \times 0.8 \times 41.75$$

$$F_{MAC} = 33.23 \text{ kmol/h}$$

Patent CN114057571A mentions the composition of the reactor outlet but the value of steam and oxygen was not given. So using the elemental balance for carbon and oxygen, the outlet flow rate of steam and water was also calculated. Nitrogen is an inert gas so the moles of nitrogen in and moles out will be the same. Table 19 gives the outlet of the reactor 1.

Table 19: Stream Composition of outlet of reactor 201

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	wt%
IB	0.03	0.17	56.00	3.61	0.03

MAC	6.84	33.23	70.00	2,326.31	16.13
CO	1.05	5.12	28.00	143.46	0.99
MAA	0.30	1.45	86.00	125.04	0.87
Acetic acid	0.30	1.45	60.00	87.24	0.60
Acetone	0.17	0.83	58.00	48.19	0.33
Acetalde	0.30	1.45	44.00	63.97	0.44
Acrolein	0.06	0.28	56.00	15.51	0.11
Formaldehyde	0.04	0.21	30.00	6.23	0.04
Acrylic acid	0.04	0.21	72.00	14.95	0.10
O2	4.99	24.23	32.00	775.44	5.38
N2	60.91	296.05	28.00	8,289.27	57.46
CO2	0.87	4.24	44.00	186.55	1.29
Steam	22.39	108.82	18.00	1,958.78	13.58
Formic acid	1.70	8.25	46.00	379.55	2.63
Total	100.00	486.00		14,426.52	100.00

Similarly, for the second oxidation reactor **R202**, the conversion of methacrolein is 78 % and the selectivity to methacrylic acid is 85 %. So the molar flow rate of methacrylic acid FMAA will be given by -

$$F_{MAA} = 0.85 \times 0.78 \times F_{MAC}$$

$$F_{MAA} = 0.85 \times 0.78 \times 33.23$$

$$F_{MAA} = 23.81 \text{ kmol/h}$$

Again using the compositions given in the patent, the outlet of the reactor R202 was also calculated. The makeup air that was to be added to the reactor, was calculated by elemental balance of Nitrogen, Oxygen and Carbon. The stream composition of the reactor outlet is given below. The mass flow rate of the makeup air is 13,127.27 kg/h.

Table 20: Stream composition of outlet of Reactor R202

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	wt%
MAC	0.70	7.31	70.00	511.79	1.86
CO	1.00	10.44	28.00	292.45	1.06
MAA	2.28	23.81	86.00	2,047.99	7.43
Acetic ac	0.31	3.24	60.00	194.27	0.71
acetone	0.12	1.25	58.00	72.69	0.26
acetaldehyde	0.21	2.19	44.00	96.51	0.35
acrolein	0.04	0.42	56.00	23.40	0.08
formaldehyde	0.03	0.31	30.00	9.40	0.03
acrylic acid	0.03	0.31	72.00	22.56	0.08

oxygen	6.50	67.89	32.00	2,172.49	7.88
steam	18.98	198.28	18.00	3,569.02	12.95
CO ₂	0.69	7.25	44.00	318.83	1.16
N ₂	62.29	650.59	28.00	18,216.55	66.11
Impurity	0.003	0.035	166.00	5.85	0.02
Total		1,044.47		27,553.80	

6.2 Quencher

Before sending the outlet gas from the reactor to the quencher, it is cooled to 230 °C and then sent to the quencher. The quenching occurs at 80 °C, this separates methacrolein, acrolein and the lighter carbon compounds in the gas phase while the liquid phase will contain some amount of Methacrolein dissolved. The liquid phase will also contain Methacrylic acid, Methacrolein, Impurity which is Terephthalic acid, acrylic acid, Acetic acid and large quantity of water. For all the separation steps after the reactor, 1000 ppm of a polymerisation inhibitor (in our case MEHQ) needs to be sprayed throughout the columns to avoid the formation of polymers. The gas stream composition is given in Table 21. The gas stream is sent to the flare for combustion.

Table 21: Gas stream composition (outlet of quencher)

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate(kg/h)	Wt%
MAC	0.96	7.14	70.00	500.09	2.30
CO	1.40	10.44	28.00	292.45	1.35

Acetone	0.17	1.25	58.00	72.69	0.33
acetaldehyde	0.29	2.19	44.00	96.51	0.44
acrolein	0.06	0.42	56.00	23.40	0.11
oxygen	9.08	67.89	32.00	2,172.49	10.01
formaldehyde	0.04	0.31	30.00	9.40	0.04
CO2	0.97	7.25	44.00	318.83	1.47
N2	87.04	650.59	28.00	18,216.55	83.94
Total		747.49		21,702.40	

Similarly the composition of the liquid stream is given in

Table 22: Liquid stream composition (outlet of quencher)

Species	Mol%	Molar flow rate (kmol/h)	MW	Mass flow rate (kg/h)	Wt%
MAA	10.54	23.81	86.00	2,047.99	35.00
MAC	0.07	0.17	70.00	11.70	0.20
Impurity	0.02	0.04	166.00	5.85	0.10

acrylic acid	0.14	0.31	72.00	22.56	0.39
acetic acid	1.43	3.24	60.00	194.27	3.32
water	87.79	198.28	18.00	3,569.02	60.99
Total		225.85		5,851.40	100.00

The liquid stream is then sent to the crystallizer to crystal out TPA at 13 °C. The entire TPA is crystallised out and the liquid stream outlet of the crystallizer is sent to an extraction column.

6.3 Extraction column

Since the compounds have the ability to polymerize, there is no data available for the Liquid-liquid equilibrium data. So for the extraction column, a simulation was run on DWSIM where the interaction parameters were calculated using UNIFAC and the model used for LLE was NRTL. Solvent chosen for extraction was n-Butyl Acetate (NBA) which has the tendency to go into the organic phase and has low solubility in water. The simulation report is attached in Appendix A.

The amount of solvent requires was 8000 kg/h. Table 23 is the stream composition of the extract phase.

Table 23: Stream composition of extract phase

Species	Mass flow rate (kg/h)	MW	Molar flow rate (kmol/h)	wt%	mol%
MAA	2,028.52	86.00	23.59	18.40	15.82
NBA	7,800.10	116	67.24	70.75	45.11
Water	986.65	18	54.81	8.95	36.77

Acrylic acid	22.57	72.00	0.31	0.20	0.21
Acetic acid	186.54	60.00	3.11	1.69	2.09
MAC	11.70	70.00	0.17	0.11	0.11
Total	11,024.38		149.07		

We can see that most of the methacrylic acid is extracted along with the other organics and some amount of water. However the solvent extracted is also almost equal to the solvent fed. This will make the distillation of this stream to purify the methacrylic acid more rigorous and energy intensive.

Table 24: Stream composition of the raffinate phase

Species	Mass flow rate (kg/h)	MW	Molar flow rate (kmol/h)	wt%	mol%	
MAA	19.47	86.00		0.23	0.69	0.15
NBA	199.90	86		2.32	7.12	1.59
Water	2,582.37	18		143.47	91.92	98.17
Acrylic acid	0.00	72.00		0.00	0.00	0.00
Acetic acid	7.73	60.00		0.13	0.28	0.09
MAC	0.00	70.00		0.00	0.00	0.00
Total	2,809.47			146.14		

From Table 24, we can see that the distribution coefficient is 27. The raffinate stream is also almost water and can be sent to the ETP.

Chapter 7: Energy balance

The values of specific heat and the enthalpy of formation were taken from Yaws' Handbook of physical Properties. Both were functions of temperature. Based on these values, energy balance was done for all the three reactors.

The enthalpy of formation (H_f) is given by

$$H_f = A + BT + CT^2,$$

where H_f is in kJ/mol, T is in K and the component is in gas phase.

Table 25: Regression coefficients for Enthalpy of Formation based on Temperature

Species	A	B	C
CO	-112.19	8.12E-03	-8.04E-06
CO2	-393.422	1.50E-04	-1.39E-06
Formaldehyde	-109.671	-2.32E-02	7.87E-06
acetic acid	-422.584	-4.84E+00	2.34E-05
Acetaldehyde	-154.122	-4.72E-02	2.03E-05
Acrolein	-66.702	-5.95E-02	3.87E-05
Acrylic acid	-325.038	-4.41E-02	2.09E-05
Acetone	-199.175	-7.15E-02	3.25E-05
Methacrolein	-90.882	-8.68E-02	5.28E-05
Methacrylic acid	-430.04	-4.75E-02	3.47E-04
Impurity (TPA)	-688.693	-1.13E-01	4.93E-05
Water	-238.41	-1.23E-02	2.77E-06
Ethylene Oxide	-38.88	-5.40E-02	2.56E-05
Isobutylene	4.126	-8.16E-02	3.66E-05

Similarly the values of C_p were calculated for the component in gas phase

$$C_p = A + BT + CT^2 + DT^3 + ET^4$$

where C_p is in J/molK and T is in K.

Table 26: Regression coefficients for Specific heat based on Temperature

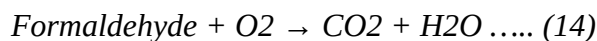
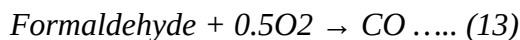
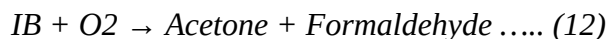
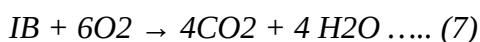
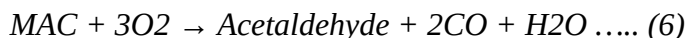
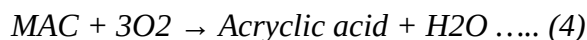
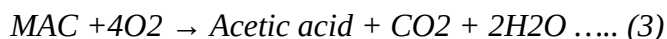
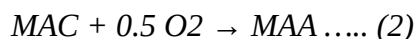
Species	A	B	C	D	E
MAL	14.506	1.59E-01	3.12E-04	-4.21E-07	1.42E-10
CO	29.556	-6.58E-03	2.01E-05	-1.22E-08	2.67E-12
MAA	-28.131	5.47E-01	-5.39E-04	2.86E-07	-6.09E-11
Acetic ac	34.85	3.76E-02	2.83E-04	-3.08E-07	9.26E-11
acetone	35.918	9.39E-02	1.87E-04	-2.16E-07	6.32E-11
acetaldehyd e	34.14	4.00E-02	1.56E-04	1.64E-07	4.72E-11
acrolein	109.243	-5.10E-01	1.71E-03	-1.81E-06	6.60E-10
formaldehy de	34.428	-2.98E-02	1.51E-04	-1.27E-07	3.39E-11
acrylic acid	7.755	2.94E-01	-2.09E-04	7.16E-08	-9.10E-12
oxygen	31.32234	-2.02E-02	5.79E-05	-3.65E-08	7.37E-15
steam	30.092	6.83E-03	6.79E-06	-2.53E-09	8.21E-14
CO2	27.437	4.23E-02	-1.96E-05	4.00E-09	-2.99E-13

Chapter 8: Equipment Sizing

For the equipments given in Table 18, the complete mechanical design was done for Reactor R201 and the extraction column T204.

8.1 Reactor R201

The feed to the reactor is stream 4, containing Isobutene (IB), steam and air entering at 350 °C. The reactor is a multi-tubular packed bed reactor, where the catalyst used is a MoVBiCsRh oxide. The outlet contains unreacted Isobutene, Methacrolein (MAC), methacrylic acid (MAA), acetone, formaldehyde, acetaldehyde, acetic acid, acrylic acid, acrolein, carbon dioxide, carbon monoxide, nitrogen, oxygen and steam. Based on this, the following reactions can be said to have occurred in the reactor;



8.1.1 Extent of Reactions

Based on the known outlet and inlet, to determine the actual reactions occurring in the reactor, for all the reactions, the extent of reactions was determined where extent of reactions ζ is given by -

where, v_{ij} is the stoichiometric coefficient of species “i” in reaction “j”. For each of the 14 reactions, the extent was calculated using the python code mentioned in Annexure 1.

From the computational calculations, the following extent of reactions were obtained

Table 27: Extent of the reactions

number	reaction	extent of the reaction
1	$\text{IB} + \text{O}_2 \rightarrow \text{MAC} + \text{H}_2\text{O}$	38.7000
2	$\text{IB} + 1.5 \text{O}_2 \rightarrow \text{MAA} + \text{H}_2\text{O}$	1.5191
3	$\text{IB} + 5\text{O}_2 \rightarrow \text{Acetic acid} + \text{CO}_2 + 3\text{H}_2\text{O}$	1.1094
4	$\text{IB} + 3\text{O}_2 \rightarrow \text{Acrylic acid} + 2\text{H}_2\text{O}$	0.0303
5	$\text{IB} + 2.5\text{O}_2 \rightarrow \text{Acrolein} + \text{CO}_2 + 2 \text{H}_2\text{O}$	0.0000
6	$\text{IB} + 3\text{O}_2 \rightarrow \text{Acetaldehyde} + 2\text{CO} + 2\text{H}_2\text{O}$	1.1094
7	$\text{IB} + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 4 \text{H}_2\text{O}$	1.1650
8	$\text{IB} + 4\text{O}_2 \rightarrow 4\text{CO} + 4 \text{H}_2\text{O}$	0.6542
9	$\text{IB} + 2.5 \text{O}_2 \rightarrow \text{Acroelin} + \text{CO}_2 + 2\text{H}_2\text{O}$	0.5194
10	$\text{MAC} + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 3\text{H}_2\text{O}$	0.0000
11	$\text{MAC} + 3\text{O}_2 \rightarrow 4\text{CO} + 3\text{H}_2\text{O}$	0.0000
12	$\text{IB} + \text{O}_2 \rightarrow \text{Acetone} + \text{Formaldehyde}$	0.6115
13	$\text{Formaldehyde} + 0.5\text{O}_2 \rightarrow \text{co}$	0.5718
14	$\text{Formaldehyde} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$	0.0000

So we can ignore reaction numbers 5, 10, 11 and 14 for the reactor design part.

8.1.2 Mass balance across the reactor

To have a accurate mass balance, it is necessary to obtain the flow rates of the products in terms of the inlet flow rates, selectivity and the conversion. So if the inlet flow rate of the limiting reactant (for our case Isobutene) is given by F_{IB0} , for oxygen it is given by F_{O20} , for nitrogen it is given by F_{inerts} and for steam it is given by F_{H2O0} , the selectivities for methacrolein, side products, carbon monoxide and carbon dioxide are Y_a , Y_b , Y_c and Y_d respectively and the conversion of isobutene is given by X_a , the mass balance can be obtained in the table 28. Here Y_a is 0.82, Y_b is 0.1, Y_c is 0.03 and Y_d is 0.05. Since the reactor is continuous and the rate equations will use the partial pressures and concentrations, this is a necessary step to do.

Table 28: Mass balance across the reactor

Species	initial	At any point in the pbr
MAC	0	$F_{IB0} * X_a * Y_a$
Side products	0	$F_{IB0} * X_a * Y_b$
CO	0	$F_{IB0} * X_a * Y_c * 4$
CO2	0	$F_{IB0} * X_a * Y_d * 4$
IB	F_{IB0}	$F_{IB0} * (1 - X_a)$
O2	F_{O20}	$F_{O20} - (F_{IB0} * X_a * Y_a - F_{IB0} * 4.75 * x_a * Y_b - F_{IB0} * X_a * Y_c - F_{IB0} * X_a * Y_d * 6/4)$
INERTS	F_{inerts}	F_{inerts}
H2O	F_{H2O0}	$F_{H2O0} + F_{IB0} * X_a * Y_a + F_{IB0} * X_a * Y_b + F_{IB0} * X_a * Y_c * 4 + F_{IB0} * X_a * Y_d * 4$
Total	F_{T0}	$F_{T0} + F_{IB0} X_a (6 Y_c + 5.5 Y_d - 3.75 Y_b)$

8.1.3 Process design

The values of the partial pressures and the concentrations to be used in the kinetic rate equations will be substituted as the function of the flow rates, selectivities and conversions as given in table 28. Then the design equation for a packed bed reactor was used to give the following ode which was then solved for in Python (annexure 2).

Here $-R_{ib}$ is the total consumption of isobutene for all the reactions where it is a reactant.

Solving this ode, the weight of the catalyst is obtained as 21618.89 kg.

8.1.4 Catalyst properties

The particle density of the catalyst is said to be 2400 kg/m^3 and the voidage is 21%. The specific area of the catalyst is $3.4 \text{ m}^2/\text{g}$ and the diameter is 30 mm.

8.1.5 Mechanical design

Using the weight of the catalyst and the bulk density, we obtain the volume of the bed as 11.38 m^3 . Assuming a fixed height to diameter or **H/D** ratio for the catalyst bed, we can find the Height of the bed that will be required. Assuming H/D to be 8 (patents say the H/D ratio has the range of 3-20), we get the Height of the bed as 9.75 m. Assuming the support space as 15%, the height of the cylinder can be found:

$$H_{\text{bed}} \times (1 + 0.15) = 9.75 (1.15) = 11.21 \text{ m.}$$

For the safe design, the design pressure and temperature should be 1.1 and 1.5 time the operating pressure and temperature. Hence the design pressure is 2.42 bar and design temperature will be 934.73 K.

The volumetric flow rate of the inlet is calculated by using ideal gas law and it comes out to be $3.17 \text{ m}^3/\text{s}$. Hence the space time for the reactor is the bed volume divided by the volumetric feed flow rate which comes to be 4.13 sec.

Reactor Sizing Parameters					
$V_{\text{bed}} (\text{m}^3)$	$\text{Diameter} (\text{m})$	$H_{\text{bed}} (\text{m})$	$H_{\text{cylinder}} (\text{m})$	$A_{\text{cross}} (\text{m}^2)$	$V_{\text{cylinder}} (\text{m}^3)$
11.38	1.22	9.75	11.21	1.17	13.09

8.1.5.1 Materials of construction

Understanding the performance of different materials is one of the most essential parameters when designing a reactor vessel. For safe and reliable chemical operations, materials' long-term behavior in reactor core, ability to resist degradation and corrosion must be taken into consideration. Furthermore, the materials of construction must be economical, meet mechanical requirements at specific process

conditions and allow the maintenance and replacement. Depending on the situation, carbon steel and stainless steel are most practical and convenient in terms of both mechanical properties and financial budget. However, bismuth molybdate catalyst and water that is utilized for the reaction might cause severe corrosion for carbon steel material, which is why stainless steel is preferred for the reactor construction. Also, stainless steel is satisfactory for machining, welding, hot and cold working, and it has significant thermal resistance and tensile strength. Numerous types of stainless steel are available, and they are generally categorized into three classes, Ferritic, Austenitic and Martensitic. The uniform structure of Austenitic steel provides remarkable corrosion resistance, which is the main reason why it has widely been utilized in process industries till now. Therefore, the reactor vessel is constructed by Type 304(18/8) Austenitic stainless steel with minimum chromium and nickel content (stable structure).

BS1501	801B
AISI	304
Composition	0.08% C, 2% Mn, 8 – 11% Ni, 17.5 – 20% Cr,
Thermal Conductivity (W/m°C)	16.2
Melting Point(°C)	1400 – 1455
Density (kg/m ³)	8000
Tensile Strength at 380°C (MPa)	114

8.1.5.2 Tube design

Assuming a tube outer diameter of 31.75, length 12 m and a trial thickness of 2mm, we can calculate the thickness using the thickness for a vessel under pressure. We calculate the ratio of outer diameter to assumed thickness and the ratio of length to the outer diameter to get the value of factor A and factor B to get the allowable pressure value. If allowable pressure is more than design pressure, the thickness will be adequate. The following results are obtained from the calculations -

t _{tube_trial}	2
Do/t	15.88
L/Do	353.19
Factor A	5.00E-05
Factor B	5.50
Allowable P (mPa)	0.46

Hence the allowable pressure is 4.6 bar which is more than our design pressure.

Now to get the number of tubes, we calculate the inner volume of the tube using the formula $V = \pi r_i^2 L$ which comes to be 7.26 mm³. The volume of the catalyst should be equal to the number of tubes into the volume of one tube. Hence we get the number of tubes as 1242.

The cross section area of one tube will be calculated as 604.81 mm². Now using this we can get the velocity in one tube using the formula

$Q = \text{Area into velocity into number of tubes}$

So velocity of gas in one tube will be 4.22 m/s.

Using the formula for baffle diameter for square pitch, we get the baffle diameter as 1.81m, if we assume baffle diameter to be 95% of shell diameter, the shell diameter is 1.91m.

8.1.5.3 Shell design

The shell side will have the heat transfer fluid. The chosen thermic fluid in this case is LiNaKNO₃ salt. It will be passed in the molten form. The melting point is 451 K. The calculations for the shell side are given below

cp_salt	1.65 kJ/kgK		
T_in	401 K	T_rise	252.37
T_out (max)	653.37 K		
velocity	0.80 m/s	range is 0.6-	
density	2700 kg/m ³	1.5 m/s	
flow rate	7.80E-02 m ³ /s	Viscosity (Pas)	
Q_req	6.90 MW	210.63 kg/s	
mcp	347.53 kW/K		
delT	19.85		

8.1.6 Heat transfer calculations

Using Dittus Boelter equation we get the heat transfer coefficient on the shell side as 211.08 W/m²K and on the tube side as 12.26 W/m²K. Using these values we design for the hotspot.

8.1.7 Accessories design

Nozzle design

Nozzles are important to control the inlet and outlet flow of the reactors and optimum diameter of the nozzles can be calculated by Harker equation:



$$D_{opt} = 8.41 * \rho^{0.31} = 8.41 * \rho^{0.45} \quad D_{opt} \text{ --optimal diameter (mm),}$$

$$w - \text{mass flow (kg/min)} \quad \frac{5273.17^{0.45}}{2.68} = 293 \text{ mm}$$

ρ – density (kg/m³).

Having calculated the optimal diameter, the thickness of nozzles can also be determined:

$$2 * \sigma * \eta - 0.2 * \frac{P_{design} * D_{opt}}{P_{t_{nozzle}}} = \frac{5.5 * 10^6 * 0.293}{2 * 100 * 10^6 * 0.9 - 0.2 * 5.5 * 10^6} = 1 \text{ mm}$$

Nozzle reinforcement

Maximal allowable stress near the nozzle decreases and that is why, reinforcing pads should be welded there with 1.5 or 2 times of the thickness of nozzle diameter [37]. Therefore, assumption could be made in this range:

$$D_{reinforcement} = D_{opt} * 1.5 = 293 * 1.5 = 440 \text{ mm}$$

440 mm reinforcement should be installed for the nozzles.

Vessel

Internals

Manhole

20-inch (0.5 meter) manhole must be designed for the reactors with diameter more than one meter. Also, only one manhole is required for the single bed vessels [37]. Therefore, one 0.5 meter manhole will be used MAL reactor.

Bursting disc

Bursting discs are safety devices designed and manufactured to relieve the system pressure if the pressure of the vessel exceeds the predetermined value. Generally, minor pressure increments

can be compensated by the relief valves, however, if the system pressure overruns the 20% of maximal design pressure, then bursting discs fail and provide full hole opening for flow. Depending on the corrosivity of the fluid, variety of construction material for the bursting discs can be used. The size range of bursting disks is from 3 mm to 1.2 m and diameter of the rupture disk can be assumed as 0.5 meter [37].

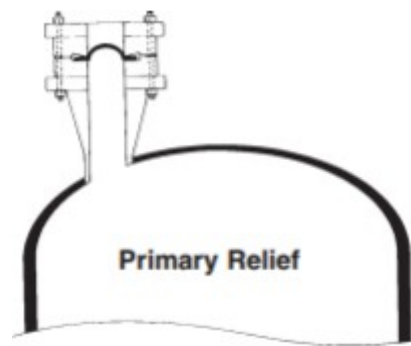


Figure 5. Bursting Disk

Wind Loads and Dynamic Wind Pressure

The load and speed of the wind should be considered when designing equipment for process plants. Therefore, the meteorological factors need to be analyzed elaborately for both onshore and offshore operations. Pressure that the wind impact to cylindrical vessels can be found by the equation below:

$$wP_w = 0.05 * u^2 \quad (2.8)$$

In which,

P_w – wind pressure (P),

u_w – wind speed (km/hr).

The wind velocity depends on the local topography and some other factors of the chosen location. Hai Pong city of Vietnam is selected as the target location for the MMA plant and maximal velocity of wind in Hai Pong is recorded as 15 km/hr in January which is the windiest month in Hai Pong [40]. However, calculations should be done for 30 km/hr for safer design.

$$P_w = 0.05 * 30^2 = 45 \text{ Pa}$$

Now, the load imposed per reactor length can be evaluated:

$$W = P_w * D_{eff} \quad (2.9)$$

W – load per unit length (N/m),

D_{eff} – effective diameter (m).

Effective diameter is calculated as the sum of reactor outside and insulation diameters, however, for isothermal reactors insulation is not installed and therefore, can be skipped.

$$W = P_w * D_{eff} = 45 * 3.265 = 147 \text{ N/m}$$

Knowing the load per unit length, it is feasible to determine the bending moment:

$$M_x = (W * H_{vessel})^2 / 2 = 12152.6 \text{ N} * m$$

M_x – maximal bending moment (N*m),

H_{vessel} – overall height of vessel (m).

Weight of Vessel

Approximate weight of the multitubular isothermal steel fixed bed reactor can be calculated by knowing the important parameters [37]:

$$W_v = 240 * C_v * D_m * (H_{vessel} + 0.8 * D_m) * t_{wall}$$

Where,

W_v – reactor weight excluding internals and fittings(kN),

C_v – vessel internals constant accounts for weight of internals (taken as 1.15),

D_m – effective mean diameter, which is the sum of internal diameter and wall thickness (m).

$$D_m = D_{internal} + t_{wall} = 3.24 + 0.014 = 3.25 \text{ m}$$

$$W_v = 240 * 1.15 * 3.25 * (12.86 + 0.8 * 3.25) * 0.014 = 194.5 \text{ kN}$$

8.2 Extraction column

The present report provides a comprehensive design of a liquid–liquid extraction (LLE) system for the recovery of Methacrylic Acid (MAA) from an aqueous feed stream using n-Butyl Acetate (NBA) as the extraction solvent. The feed stream of 6,063 kg/h contains 2,124 kg/h of MAA, and is processed in a counter-current extraction column operating at 13°C. The solvent flow rate is maintained at 10,000 kg/h to ensure favorable mass transfer and high recovery efficiency.

Based on equilibrium analysis and stage-wise calculations, the system requires 5 theoretical stages; however, considering a realistic stage efficiency of 80%, the number of actual stages is determined to be 7. The extraction column selected is a pulsed perforated plate column with a diameter of 0.9 m and a height of 6.5 m. The overall system achieves approximately 95% recovery of MAA.

Process Description

The extraction process is carried out in a counter-current mode, wherein the aqueous feed enters from the top of the column and flows downward, while the organic solvent (NBA) enters from the bottom and rises upward. Due to density differences, the aqueous phase acts as the continuous phase, and the organic phase is dispersed through pulsation.

Mass transfer of MAA occurs at the interface between the two immiscible phases, driven by concentration gradients and governed by equilibrium relationships. The process results in an extract stream rich in MAA and a raffinate stream depleted of MAA.

Thermodynamic Equilibrium

The equilibrium relationship between MAA concentration in the aqueous and organic phases is expressed as:

$$x_{org} = -0.847 x_{aq}^2 + 2.153 x_{aq} + 0.0012$$

$$Y' = 4.234 (X')^{0.892}$$

The distribution coefficient varies with concentration due to molecular interactions such as dimerization, and an average value of 3.5–4.0 is used for design purposes.

Kremser Equation and Stage Analysis

The number of theoretical stages is calculated using the Kremser equation for counter-current extraction:

$$N = \ln[((X'_f - Y'_m)/(X'_r - Y'_m)) (1 - 1/E) + 1/E] / \ln(E)$$

The extraction factor is given by:

$$E = m' (S'/F')$$

Substituting the design parameters yields an extraction factor of approximately 9.65, and the calculated number of stages assuming constant slope is 1.27. However, due to variation in equilibrium slope, stage-by-stage analysis indicates that 5 theoretical stages are required.

Stage Efficiency and Actual Stages

Considering a stage efficiency of 80%, the number of actual stages is calculated as:

$$N_{\text{real}} = N_{\text{theoretical}} / \text{efficiency} = 5 / 0.80 = 6.25 \approx 7 \text{ stages}$$

6. Column Design

The column diameter is determined based on volumetric flow rate and allowable velocity, resulting in a calculated value of 0.305 m; however, a standard diameter of 0.9 m is selected for safe operation.

The height of the column is calculated using:

$$H = N \times \text{HETS} = 5 \times 0.85 = 4.25 \text{ m}$$

Including disengagement zones and design margin, the total height is taken as 6.5 m.

7. Material Balance

The overall mass balance is satisfied with total input equal to total output. The MAA recovery is calculated as:

$$\text{Recovery} = (2017.8 / 2124.0) \times 100 = 95\%$$

8. Equipment Specifications

The extraction is performed in a pulsed perforated plate column with the following specifications:

Diameter: 0.9 m, Height: 6.5 m, Plates: 85, Plate spacing: 50 mm, Pulsation frequency: 1.5 Hz,

Amplitude: 15 mm. The material of construction is SS316L.

9. Conclusion

The designed LLE system successfully achieves 95% recovery of Methacrylic Acid under the specified operating conditions. The design is technically feasible, economically viable, and suitable for industrial implementation.

References

A.S.Huseynov, S.N.Guliyeva, “*Kinetic regularities of the catalytic oxidation of isobutylene to methylacrolein on an antimony-tin catalyst*”, AZERBAIJAN CHEMICAL JOURNAL, 1, 2023

Bhanu Pratap, Ravi Kant Gupta, Bhuvnesh Bhardwaj, Meetu Nag, “*Resin based restorative dental materials: characteristics and future perspectives*”, Japanese Dental Science Review, 2019, 55(1), 126-138

CHANGZHOU HICKORY CHEMISTRY Co Ltd, China Patent 105968011A.

CHANGZHOU TIANMA GROUP Co. Ltd., China Patent 102584579A.

Chemical Weekly, 2024- 2025 issues

Curtis Ingstad Carlson, Jr. Michael Stanley DeCourcy, Jamie Jerrick John Juliette, US Patent 7253307B1

Hoornaerts, S., Vandeputte, D., Thyron, F. C., Ruiz, P., & Delmon, B. (1997), “*Modification of kinetic parameters by action of oxygen spillover in selective oxidation of isobutene to methacrolein*”. Catalysis Today, 33(1), 139–150. [https://doi.org/10.1016/S0920-5861\(96\)00109-5](https://doi.org/10.1016/S0920-5861(96)00109-5)

Ivanova P. et. al., “Kinetics of the Esterification of Methacrylic Acid with Ethylene Oxide in the Presence of Ferric Chloride Immobilized on Polymer Support”, Applied Catalysis, 53 (1989) 41-52.

John R. Cadwell, US patent S2484487A

Montheard, J. P., Chatzopoulos, M., & Chappard, D. “*2-Hydroxyethyl Methacrylate (HEMA): Chemical Properties and Applications in Biomedical Fields*”, Journal of Macromolecular Science, Part C, 1992, 32(1), 1–34.

Roma Chemical Co Ltd, China Patent 101497563B

Roma Chemical Co Ltd, China Patent 114057571A

Sigma Aldrich, “HEMA sds”, 2006

S.N. Dobić et al., “*Synthesis and characterization of poly(2-hydroxyethyl methacrylate/itaconic acid/poly(ethylene glycol) dimethacrylate) hydrogels*”, Chemical Engineering Journal 179 (2012) 372–380

Yaws, C. L. (Ed.). (2015). “Yaws’ Handbook of Thermodynamic and Physical Properties of Chemical Compounds”. McGraw Hill.